Semantic Prototyping With CLIP for Few-Shot Object Detection in Remote Sensing Images

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***Abstract*— Few-shot object detection (FSOD) has been proposed to solve the problem of insufficient data for training, and it has drawn the attention of the remote sensing community in recent years. A mainstream type of FSOD method is to generate class prototypes based on the limited samples to help the construction of classification decision boundaries. However, these constructed prototypes may be far away from the true class centroids in the few-shot scenario. Recently, the vision-language model (VLM) has shown its powerful ability to align the visual features and text features, which leads to strong zero-shot performance on various downstream computer vision tasks when given only texts. Therefore, in this work, we propose to build class prototypes from text descriptions instead of limited visual instances by leveraging a classical pretrained VLM named CLIP. Concretely, we generate prototypes by feeding the CLIP text encoder with class names and enforcing each positive proposal feature to be close to the corresponding prototype. To accelerate the alignment process, we utilize the CLIP visual encoder as another teacher to achieve visual knowledge distillation. Moreover, we adopt prompt tuning to adapt CLIP to the remote sensing scenario. Extensive experiments on two public FSOD datasets, i.e., DIOR and NWPU VHR-10.v2, demonstrate the effectiveness of our method, which yields competitive results with that of existing approaches.**

***Index Terms*— Few-shot object detection (FSOD), remote sensing images (RSIs), vision-language model (VLM).**

I. Introduction

OBJECT detection in remote sensing images (RSIs) has attracted much attention in recent years, which aims at recognizing and locating the objects of interest in the images. As a long-standing and fundamental task of aerial and satellite image analysis, object detection has been applied to a variety of fields such as urban planning, geological hazard detection, environmental monitoring, and land-use/land-cover (LULC) mapping [1], [2]. Thanks to deep convolutional neural network (DCNN) [3], [4] and Transformer [5], [6] with strong ability to extract robust image features, advanced object detectors [7], [8], [9], [10], [11], [12], [13], [14] have made tremendous progress in natural scenes. To solve the specific problems with satellite images such as arbitrary orientation and densely packed objects, various object detection algorithms have also been proposed in the remote sensing community [15], [16], [17], [18], [19], [20], [21], [22], [23], [24]. Despite awesome performances obtained by state-of-the-art object detectors, an avalanche of annotated data is required for training [25]. However, labeling sufficient training data is labor-intensive and expensive, and it is difficult to collect enough samples for some rare object categories [26]. Consequently, it is crucial to make the detectors robust in data-scarce scenarios.

To reduce the demand for training data, few-shot learning (FSL) [27] has been proposed to emulate the human ability to swiftly understand novel concepts. Generally, FSL aims at fully exploiting knowledge from limited data without overfitting and improving the generalization ability to novel categories. Few-shot object detection (FSOD) intends to transfer previously learned knowledge from the base classes with abundant annotated samples to help the detection of novel classes where each class has only a few instances. A number of FSOD methods have been proposed [28], [29], [30], [31], [32], [33], [34], [35] in recent years for natural scenes. However, it is suboptimal to directly apply off-the-shelf generic FSOD methods to the remote sensing scenario. As compared to objects in natural scenes, objects in aerial images are more prone to spread out amongst complex backgrounds and possess higher interclass similarity but lower intraclass similarity. A key solution is to design a strong classifier to alleviate the misclassification. Some pioneering works [36], [37], [38], [39] adopt a prototype-based framework [40], which generates class prototypes from the limited data and aggregate them with object features to make the model easier to be classified. The intuition behind this framework is that objects belonging to the same class should have similar features, and thereby they are close to the corresponding prototype. However, due to high intraclass diversity, there may be a huge gap between the prototype generated from limited instances and the true class centroid. Moreover, instances far from the prototype may be misclassified as other classes.

Compared with varied visual information in different images, semantic information for each category in texts is relatively invariant and unrelated to data availability, which

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is more robust for constructing class prototypes in the few-shot scenario. Meanwhile, VLMs have shown strong ability for multimodal alignment due to large-scale pretraining, which makes it easier to compare visual features and text features in the same feature space. Therefore, we propose a new FSOD method for RSIs that transfers the knowledge from a classical pretrained VLM to generate more reliable class prototypes. Specifically, based on the generic two-stage fine-tuning FSOD architecture, we employ CLIP [41], the abbreviation for contrastive language image pretraining, as an auxiliary prototype generator, which is pretrained on 400 million image–text pairs in a self-supervision way and has shown its generalization ability in various computer vision tasks. After obtaining the prototype for each class by feeding class names with a prompt template into CLIP's text encoder, we optimize proposal features extracted by the object detection model via the cross-entropy loss, treated as language knowledge distillation. As proposal features and class prototypes described by natural language are in different feature spaces, we leverage the frozen visual encoder of CLIP as a teacher model to achieve feature alignment. Since CLIP is trained to extract image-level representations to match the input texts, which is not suitable for object detection, we crop each image into multiple patches corresponding to the ground-truth boxes with scale extensions and then feed them into the visual encoder of CLIP as target instance-level features. Furthermore, we apply prompt tuning [42] to achieve domain adaptation for CLIP as it is pretrained mostly on the natural image–text pairs.

In summary, our major contributions are as follows.

1) We employ the pretrained vision-language model (VLM) CLIP to generate more reliable class prototypes from text descriptions and optimize proposal features by aligning the prototypes.
2) We leverage the CLIP visual encoder as a teacher model to achieve visual knowledge distillation.
3) We further apply prompt tuning to mitigate the domain gap between the pretrained CLIP and the remote sensing scenario.
4) Extensive experiments on two popular remote sensing FSOD datasets show that our proposed method is competitive to the existing FSOD methods, especially in low-shot settings.

It is worth noting that the proposed feature optimization modules are only applied during training, and the test phase is the same as the original baseline model. Therefore, there is no extra inference cost.

The rest of this article is organized as follows. Section II briefly reviews related work, including object detection and FSOD in both natural and remote sensing scenes. Section III introduces the task setting of FSOD and the basic framework used in our method. Our model is introduced in Section IV in detail. Section V reports the experimental results on two popular remote sensing FSOD datasets. Finally, Section VI draws the conclusion and highlights the future work.

II. Related Work

In this section, we briefly review related work from two aspects, generic object detection and FSOD.

A. Object Detection

Object detection is one of the most fundamental problems in computer vision, which requires the model not only to recognize objects but also to localize them in the image.

In the deep learning era, object detection task is decoupled as classification and regression [43]. And methods can be divided into two-stage, one-stage, and end-to-end detectors, where the former two are based on DCNN [3] while the latter one is based on Transformer [5]. Two-stage detectors first generate region proposals (region of interests, RoIs) and then fine-tune proposals in the second stage, such as R-CNN series [7], [10], [43], [44]. In contrast, one-stage detectors like YOLO series [11], [45], [46] and RetinaNet [9] directly predict class probabilities and coordinates on the image feature map densely. Both two-stage and one-stage detectors need postprocessing such as nonmaximum suppression (NMS) to eliminate redundant boxes, which is not end-to-end. DETR [13] develops a new end-to-end object detection paradigm based on the Transformer encoder–decoder architecture [5], which generates a set of object queries from the feature map and predicts class and coordinate for each of them. After that, various Transformer-based detectors [14], [47] have been proposed and achieved higher detection performance.

Compared to objects in natural scenes, objects in RSIs are generally diversely scattered and largely scale-variant. Moreover, complex backgrounds and relatively smaller objects bring more challenges. Benefiting from the development of modern object detectors, object detection in the remote sensing community has also achieved great advances. For instance, Deng et al. [15] devised a multiscale object proposal network while Zheng et al. [48] proposed a hyper-scale block with a fine-grained multiscale structure to deal with the scale variation problem. Pang et al. [17] designed a unified and self-reinforced network to improve the detection performance of tiny objects in complex backgrounds. Due to arbitrary orientation in RSIs, oriented object detection has been an active topic. Ding et al. [16] applied spatial transformations on RoIs learned from oriented box supervision to alleviate misalignment between classification and regression. Yang et al. [18] designed localization loss to address the boundary problem for rotated bounding boxes. Han et al. [19] explicitly encoded rotation equivariance and rotation invariance for more accurate aerial object detection. Wang et al. [21] replaced the original full attention in Transformer [5] blocks with a new rotated varied-size window attention to deal with both scale-variation and arbitrary orientation problems. ACFG [49] adopts a knowledge distillation strategy to reduce parameters.

However, the detectors above require large-scale datasets to obtain satisfactory results and are prone to overfit on small-scale datasets with obvious performance degradation.

B. Few-Shot Object Detection

FSOD aims to tackle the problem of detecting objects when only a few available samples exist. In general, FSOD methods can be grouped into two kinds, i.e., methods based on meta-learning or based on fine-tuning (also called transfer learning). Models based on meta-learning train a meta-learner

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on *episodes* which are sampled from base classes to explicitly construct the few-shot scenario. For instance, FSRW [29] takes YOLO-v2 [45] as a meta-feature extractor and designs a feature reweighting module to refine the query features with the help of labeled data. RepMet [30] proposes a novel metric learning framework for FSOD. Zhang et al. [34] devised a mutual guidance mechanism for support and query features. Algorithms based on fine-tuning first train the model on the base set and then fine-tune parameters on the novel set. TFA [31] presents a simple yet effective framework without any model structure modifications to the general object detection framework, which only freezes specific parameters at the fine-tuning stage. FSCE [32] incorporates supervised contrastive learning [50] to alleviate misclassification. DeFRCN [33] decouples class-specific and class-agnostic parameters and utilizes an independent prototype generation module to improve classification accuracy. Yang et al. [35] presented an efficient pretrain-transfer framework and devised knowledge inheritance to initialize novel weights from the base weights.

Along with the development of generic FSOD methods, there are several FSOD algorithms focusing on RSIs. FSODM [51] builds a meta-learning framework based on FSRW [29], while DH-FSDet [52] extends TFA [31] by a new annotation sampling strategy and data preprocessing method with double head predictor structure. PAMS-Det [53] integrates involution operation into the backbone and a path-aggregation module into TFA for better localization accuracy. G-FSDet [39] proposes a novel loss to learn a more discriminative classifier and a representation compensation module to suppress catastrophic forgetting on the base classes. P-CNN [36] designs a prototype learning network to guide proposal generation, while MM-RCNN [37] replaces prototypes with a meta-memory module that stores the learned knowledge of each category. Based on meta-learning, MINN [54] designs a novel feature interaction strategy between the query and support images to alleviate feature misalignment and attention bias.

Different from these methods above, we adopt the pretrained VLM CLIP [41] as the prototype generator to optimize RoI feature extraction for better classification accuracy. Moreover, we further employ supervised contrastive learning [50] to obtain more representative class-specific features.

III. Preliminaries

A. Problem Setting

Following FSOD setting in FSRW [29] and TFA [31], there is a base set D_{base} consisting of base classes C_{base} with abundant annotated images for each class, and a novel set D_{novel} containing novel classes C_{novel} with only a few labeled instances for each class. It is worth noting that base classes and novel classes are nonoverlapping, i.e., $C_{\text{base}} \cap C_{\text{novel}} = \emptyset$. Generally, an N-way K-shot problem is considered, where N indicates the number of novel classes and K represents the number of labeled instances for each class. Namely, $D_{\text{novel}} = \{\{(x_i, c_i, b_i)\}_{i=1}^{K}, c_i \in C_{\text{novel}}\}_{j=1}^{N}$, where x_i indicates the object instance and c_i is the corresponding class label. b_i represents the box annotation of x_i in the image. During testing, the goal is to improve the performances on novel classes while maintaining the performances on base classes.

B. Two-Stage Fine-Tuning Framework

In this framework, there are two stages during training, including the base training stage and the fine-tuning stage. In the base training stage, the model is trained on the base set D_{base} with abundant annotations for base classes. In the fine-tuning stage, parameters are initialized from the former stage except for the box predictor and then fine-tuned on both the base set and the novel set with some parameters frozen. It is worth noting that the parameters of the two stages are not necessary to be exactly matched. To balance the number of instances of base classes and novel classes, K instances are sampled from the base set for each base class during the fine-tuning stage. The simple two-stage framework transfers the knowledge from the base set to the novel set and has shown promising ability in the FSOD task.

C. CLIP

CLIP [41] is a pretrained VLM, which was pretrained on 400 million image–text pairs to learn robust representations for downstream computer vision tasks such as action recognition and geo-localization. The key advantages of vision-language pretraining (VLP) are zero-shot capability and modularity, which means that once pretrained, the model serves as a foundation model and can be fine-tuned or directly utilized for other common vision-language tasks. CLIP has shown strong zero-shot capability on a wide range of computer vision tasks. Given a piece of text, also called the prompt, CLIP is able to decide whether the input image matches the text. Consequently, we utilize CLIP as a teacher to guide the object feature learning of the detection model via knowledge distillation. As the background is complex in RSIs which is hard to describe as certain prompts, we treat CLIP as an auxiliary module that only optimizes the foreground object features instead of directly applying it as the foundation model.

IV. Proposed Method

The overall architecture of our model is shown in Fig. 1. We adopt TFA [31] as the baseline, which takes Faster R-CNN [7] as the base model. It consists of several components, i.e., the backbone (including ResNet [3] and FPN [8]) that extracts visual features from the input images, the region proposal network (RPN) that generates proposals, the RoI Align module that projects proposals onto the image feature map, and the prediction head that achieves both object classification and regression. The training framework follows the two-stage fine-tuning strategy as TFA [31]. Based on it, we propose to optimize RoI features during the fine-tuning stage guided by the text prototypes with the help of a classical pretrained VLM called CLIP [41]. Specifically, we add an auxiliary module called CLIP prototype head to the original prediction head, learning more robust instance features.

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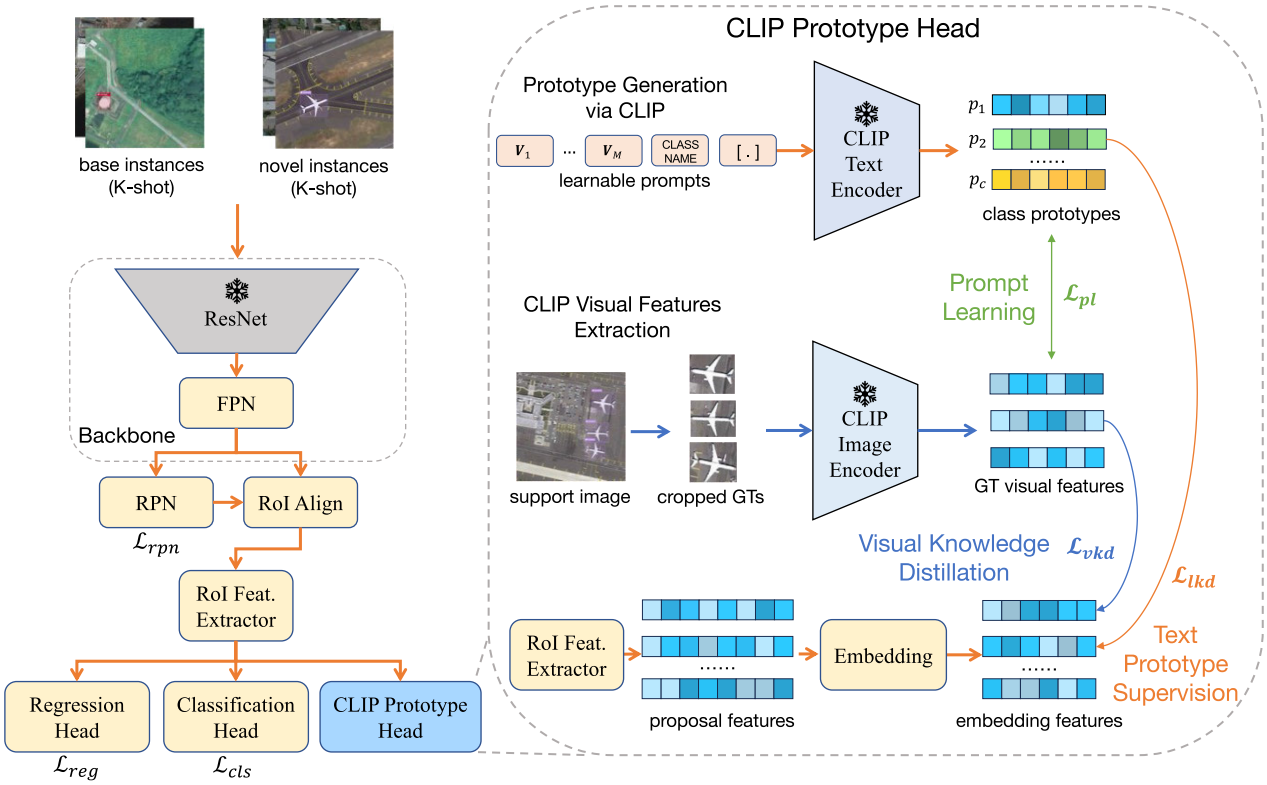


Fig. 1. Architecture of the proposed method TPG-FSOD. The training process follows the two-stage fine-tuning framework. The base training stage is not illustrated for simplicity. In the fine-tuning stage, only ResNet [3] in the backbone is frozen while other components are fine-tuned. More importantly, we propose to utilize text prototypes extracted from a pretrained VLM named CLIP [41] to optimize the proposal features. We import prompt learning to better adapt CLIP to RSIs as it is pretrained mostly on natural images–texts pairs. Moreover, we utilize the CLIP visual encoder as a teacher to accelerate the alignment between text prototypes and proposal features.

A. Training Framework

The training framework consists of two stages, i.e., the base training stage and the fine-tuning stage.

1) Base Training Stage: In the first stage, the model is trained on the base set with sufficient annotations, which is exactly the same as traditional object detection models. The joint loss is as follows:

$$\mathcal{L}_{\text{det}} = \mathcal{L}_{\text{rpn}} + \mathcal{L}_{\text{cls}} + \mathcal{L}_{\text{reg}} \tag{1}$$

where $\mathcal{L}_{\text{rpn}}$ denotes the loss of RPN [7] that learns to distinguish objects from backgrounds and coarsely regress the proposals. $\mathcal{L}_{\text{cls}}$ is the loss of the classifier, which is set as the cross entropy loss as regular object detection methods, while $\mathcal{L}_{\text{reg}}$ represents the regression loss, which is a smoothed L_1 loss proposed by Faster R-CNN [7].

2) Improved Fine-Tuning Stage: In the second stage, the model parameters are inherited from the first stage and fine-tuned on both base classes and novel classes. To balance the number of instances between base classes and novel classes, K instances of each base class are randomly sampled as training data. In this stage, TFA [31] freezes all parameters except for that of the prediction head. However, this strategy limits the performance on novel classes as it basically relies on the general knowledge learned from base classes. By comparing different parameter fine-tuning strategies, it is observed that freezing solely the parameters of ResNet [3] within the backbone yields the best tradeoff between base classes and novel classes. Concrete analysis is presented in the ablation study later.

Moreover, TFA [31] replaces the original fully connected layer-based classifier with the cosine similarity-based classifier, in which the instance-level feature normalization is used to reduce intraclass variance and maintain the performance on base classes. The cosine similarities between the input x_i and the weight vector w of the classifier are calculated as follows:

$$s_{i,j} = \alpha \frac{x_i \cdot w_j^\top}{\|x_i\| \cdot \|w_j\|} \tag{2}$$

where j indicates the class index of the weight vector, and α is a scaling factor. We set α as 20 empirically in our experiments. The predicted class scores are obtained from the cosine similarities as follows:

$$p_{i,c} = \frac{e^{s_{i,c}}}{\sum_{j=0}^{C} e^{s_{i,j}}} \tag{3}$$

where C is the total number of classes and $j = 0$ means the background class.

B. Text Prototype Supervision via CLIP

The idea of prototypical networks [40] has been widely adopted in FSOD [36], [37], [55], [56], which is built upon the concept that objects of the same class represent similar features and thus they are closer to the corresponding class prototype than other prototypes. It is helpful for reducing

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misclassification by aggregating RoI features with prototypes in an attentive manner based on the cosine similarities. However, scarce samples are hard to represent the whole class data distribution, and prototypes generated from them are unreliable, limiting the performance.

Recently, pretrained VLMs have shown strong zero-shot capability in multiple tasks and generate visual and text features in a multimodal feature space. Therefore, it is helpful to transfer the knowledge contained in VLMs into FSOD by extracting text features from class names without the need for extra training data. Specifically, we apply a representative method CLIP [41] to generate text class prototypes. Given the cth class name cls, the prompt template T_c is designed as the simplest one as CLIP: "a centered satellite photo of [cls]," and the class prototype z_c^t is then generated by feeding the CLIP text encoder with the prompt T_c.

We first introduce a nonlinear embedding head that is parallel to the original box predictor, consisting of two linear layers and an activation function rectified linear unit (ReLU) between the two layers, to project each RoI feature x_i into a vector z_i with the same dimension as CLIP output features. Subsequently, we measure the cosine similarities between z_i and class prototypes to estimate the matching scores as follows:

$$s_{i,c} = \alpha \frac{z_i \cdot {z_c^t}^\top}{\|z_i\| \cdot \|z_c^t\|}$$
$$\boldsymbol{s}_i = \left[s_{i,1}, s_{i,2}, \ldots, s_{i,C}\right] \tag{4}$$

where α indicates the scaling factor and C denotes the number of classes. We set α as 20 empirically in our experiments. As the background is hard to describe as a single feature vector, it is worth noting that only features of proposals assigned with a foreground ground-truth box are fed into the CLIP prototype head.

The objective function is finally formulated as follows:

$$\mathcal{L}_{\text{lkd}} = \frac{1}{N_{\text{pos}}} \sum_{i=1}^{N} \mathbb{1}\left\{y_i \neq C_{\text{bg}}\right\} \cdot \mathcal{H}(\text{softmax}(\boldsymbol{s}_i), y_i) \tag{5}$$

where N represents the number of proposals in each image. $\mathbb{1}\{\cdot\}$ is the indicator function and y_i denotes the class label of z_i. $\mathcal{H}$ represents the cross entropy loss.

C. Visual Knowledge Distillation via CLIP

Since RoI features extracted by the detection model are in different feature spaces with visual features from the CLIP visual encoder that may impede the prototype matching, we propose to utilize the CLIP visual encoder as the teacher to guide RoI features optimization inspired by ViLD [57].

The CLIP visual encoder was pretrained to extract image-level features, thereby we generate target visual features by feeding the encoder with cropped patches of the original image I according to the ground truth box annotations $\boldsymbol{b}$. Objects in RSIs are prone to be small, and the textures of some objects are simple, thereby we extend $\boldsymbol{b}$ to obtain some context information for better visual features. The target visual features are finally generated as follows:

$$z_i^v = \xi_v(\text{Crop}(I, \text{Scale}(b_i))). \tag{6}$$

The parameters of the CLIP visual encoder ξ_v are frozen during training. Then the visual knowledge distillation is formulated as the L1 loss between each embedded ground-truth feature z_i and its target visual feature z_i^v as follows:

$$\mathcal{L}_{\text{vkd}} = \frac{1}{M} \sum_{i=1}^{M} \|z_i - z_i^v\|_1 \tag{7}$$

where M denotes the number of ground-truth boxes in the image I.

After optimization guided by both the text encoder and visual encoder of CLIP, the detection model is able to learn a robust feature space for each class, which is helpful for improving classification performance.

D. Domain Adaptation via Prompt Learning

CLIP was pretrained on image–text pairs collected from the Internet, most of which are natural images. Since the distribution of natural images and RSIs are very different, there exists a domain gap for directly applying CLIP for classifying instances in the RSIs, which means the outputs from the text encoder and the image encoder may not match correctly. A simple way to bridge the domain gap is to fine-tune CLIP on the RSIs, but it is inefficient and carries a high risk of overfitting.

It is revealed by researchers that the prompt fed into the VLM plays a key role in downstream tasks, and prompt learning [42], [58], [59] was proposed to dynamically adapt the pretrained model to different tasks. Here we adopt the idea from CoOp [42] which replaces the fixed prompt with learnable vectors and freezes all parameters of CLIP to alleviate the domain gap problem.

The prompt of the cth class is given as follows:

$$T_c = [\mathbf{V}]_1[\mathbf{V}]_2 \ldots [\mathbf{V}]_M[\text{cls}] \tag{8}$$

where each $[\mathbf{V}]_m$ reprents a learnable vector and [cls] denotes the token embedding of the cth class name from CLIP. M is the hyperparameter to control the number of prompt vectors. It is worth noting that all prompt vectors are shared among different classes. The representation of each prompt from the frozen CLIP text encoder is still taken from the [EOS] token position. We still utilize contrastive learning to align the class prompt features with corresponding support visual features from the frozen CLIP image encoder. Given a support instance visual feature z_i^v, the probability of matching the cth class prompt is calculated as follows:

$$p_{i,c} = \frac{\exp\left(z_i^v \cdot \xi_t(T_c)^\top / \tau\right)}{\sum_k^C \exp\left(z_i^v \cdot \xi_t(T_k)^\top / \tau\right)} \tag{9}$$

where ξ_t denotes the CLIP text encoder and τ is the temperature coefficient controling the probability distribution shape. Then the objective function is defined as follows:

$$\mathcal{L}_{\text{pl}} = \frac{1}{N_{\text{support}}} \sum_{i}^{N_{\text{support}}} \mathcal{H}(p_i, y_i) \tag{10}$$

where N_{support} is the number of ground-truth boxes in the batch.

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TABLE I
DIFFERENT BASE/NOVEL CLASS SPLITS OF DIOR [60] AND NWPU VHR-10.V2 [61]

dataset	split	novel classes	base classes
DIOR [60]	1	baseball field, basketball court, bridge, chimney, ship	others
	2	airplane, airport, expressway toll station, harbor, ground track field	others
	3	dam, golf field, storage tank, tennis court, vehicle	others
	4	expressway service area, overpass, stadium, train station, windmill	others
NWPU VHR-10.v2 [61]	1	airplane, baseball diamond, tennis court	others
	2	basketball court, ground track field, vehicle	others

E. Loss Function

The three parts proposed by our method are all based on the embedding head parallel to the prediction head. The overall loss function in the fine-tuning stage can be written as follows:

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \beta \cdot \mathcal{L}_{\text{lkd}} + \gamma \cdot \mathcal{L}_{\text{vkd}} + \eta \cdot \mathcal{L}_{\text{pl}} \tag{11}$$

where β, γ, and η are hyperparameters. We simply set β and η both as 1.0 while γ as 0.1 to balance the loss values.

V. EXPERIMENTS

In this section, we conduct extensive experiments on two popular few-shot remote sensing object detection benchmarks, i.e., DIOR [60] and NWPU VHR-10.v2 [62], to evaluate the proposed method by comparing it with existing approaches. We first introduce the datasets and evaluation metrics, and then elaborate on our implementation details. Next, we compare our method with state-of-the-art FSOD methods and conduct a series of ablation studies to verify the effectiveness of each module in our model.

A. Datasets and Evaluation Metric

DIOR [60] is a large-scale optical RSI benchmark for object detection, which contains 23 463 RSIs and 20 categories with 192 472 annotated instances collected from Google Earth. The object categories include airplane, airport, baseball field, basketball court, bridge, chimney, dam, expressway service area, expressway toll station, harbor, golf course, ground track field, overpass, ship, stadium, storage tank, tennis court, train station, vehicle, and windmill. The dataset consists of three parts: the training set, the validation set, and the test set, including 5862, 5863, and 11 738 images, respectively. The image sizes are all 800 × 800 pixels in the dataset and the spatial resolutions range from 0.5 to 30 m. We follow FSODM [51] to construct the few-shot detection scenario. Specifically, there are four different base/novel splits, where each split includes 15 base classes and 5 novel classes. Table I presents the specific classes of each split. The number of shots K is set to be 3, 5, 10, 20, and 30. We combine the training and validation sets for training and adopt the test set for validation on both base and novel classes.

NWPU VHR-10.v2 [62] is another common benchmark including 1172 images of 10 categories. The object categories include airplane, baseball diamond, basketball court, bridge, ground track field, harbor, ship, storage tank, tennis court, and vehicle. The image sizes are all 800 × 800 pixels in the dataset. We follow SCoDANet [66] to split all categories into two base/novel splits detailed in Table I, where each split contains 3 novel classes and 7 base classes. The number of shots K is set to be 3, 5, 10, and 20.

Evaluation Metric: As in existing FSOD methods, mean average precision (mAP) with an IoU of 0.5 is adopted as the evaluation metric in our experiments. To better evaluate the performance on both base classes and novel classes, we report the mAP on both of them, which are denoted as bAP and nAP, respectively.

B. Implementation Details

We take TFA [31] as our baseline, which sets ResNet-101 [3] with FPN [8] as the backbone. Our model is trained on base classes for 36k iterations in the base training stage. The learning rate is initialized as 0.01 after warm-up and is divided by 10 after 24k and 32k iterations. As for the fine-tuning stage, the number of training iterations is set to 8k for all shot settings. The learning rate is set as 0.001 after the warm-up steps. In both of the two stages, we adopt SGD as the optimizer with a momentum of 0.9 and weight decay of 0.0001. The training batch size is set to 4 on 2 NVIDIA RTX 4090 GPUs. A multiscale training strategy is not applied for a fair comparison. To obtain stable results, we run each experiment three times and take the average of their results as the final performance.

C. Comparison With State-of-the-Art FSOD Methods

We compare our method with several state-of-the-art FSOD methods on the two popular datasets to demonstrate its effectiveness. The compared FSOD methods include those designed for generic FSOD tasks and those specifically tailored for RSIs. In addition, the general object detection method Faster R-CNN [7] is also included.

1) *Generic FSOD Methods*: FSRW [29] introduces a feature reweighting strategy that leverages meta-learning to focus on relevant features for novel object detection. Meta RCNN [63] also adopts meta-learning for feature refinement. Another classical FSOD method TFA [31] proposes a simple yet efficient strategy based on transfer learning. DeFRCN [33] extends the strategy of TFA [31] and integrates a decoupled fine-tuning strategy. FSDetView [67] combines feature representation learning with a viewpoint consistency mechanism.
2) *FSOD Methods Designed Specifically for RSIs*: P-CNN [36] utilizes prototype learning to improve class representation, while MM-RCNN [37] proposes a memory strategy to store the learned knowledge for

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TABLE II

COMPARISON OF FSOD DETECTION RESULTS ON DIOR DATASET UNDER 3-, 5-, 10-, 20-, AND 30-SHOT SETTINGS AND FOUR DIFFERENT BASE/NOVEL CLASS SPLITS. nAP DENOTES THE MAP ON NOVEL CLASSES WHILE bAP DENOTES THE mAP ON BASE CLASSES.† MEANS WE OBTAIN RESULTS BY DIRECTLY RUNNING THE RELEASED CODE

#Shots	Method	Class Split1		Class Split2		Class Split3		Class Split4		Average	
		nAP	bAP	nAP	bAP	nAP	bAP	nAP	bAP	nAP	bAP
3	Faster R-CNN [7]	0.000	0.459	0.000	0.457	0.000	0.510	0.000	0.484	0.000	0.478
	FSRW [29]	0.075	0.494	0.048	0.485	0.078	0.455	0.037	0.482	0.060	0.479
	Meta RCNN [63]	0.140	0.463	0.105	0.482	0.122	0.495	0.116	0.504	0.121	0.486
	P-CNN [36]	0.180	0.470	0.145	0.489	0.165	0.495	0.152	0.498	0.161	0.488
	MM-RCNN [37]	0.198	0.472	**0.156**	0.487	0.167	0.504	0.164	0.517	0.171	0.495
	SAGS&TFS [64]	0.364	-	0.197	-	0.286	-	0.237	-	0.271	-
	†DeFRCN [33]	0.302	0.622	0.104	0.607	0.134	0.634	0.153	0.618	0.173	0.620
	G-FSDet [39]	0.276	0.689	0.141	0.692	0.160	0.711	0.167	0.690	0.186	0.696
	SAE-FSDet [65]	0.288	-	0.140	-	0.167	-	**0.173**	-	0.192	-
	†TFA [31]	0.138	**0.706**	0.046	**0.706**	0.094	**0.726**	0.116	**0.698**	0.099	**0.709**
	Ours	**0.342**	0.627	0.143	0.613	**0.201**	0.656	0.131	0.601	**0.203**	0.624
5	Faster R-CNN [7]	0.000	0.441	0.000	0.463	0.000	0.487	0.000	0.518	0.000	0.477
	FSRW [29]	0.121	0.497	0.070	0.468	0.137	0.479	0.068	0.485	0.099	0.482
	Meta RCNN [63]	0.168	0.482	0.125	0.502	0.154	0.500	0.136	0.512	0.146	0.499
	P-CNN [36]	0.228	0.484	0.149	0.491	0.188	0.499	0.175	0.499	0.185	0.493
	MM-RCNN [37]	0.239	0.488	0.155	0.499	0.197	0.495	0.187	0.507	0.195	0.497
	SAGS&TFS [64]	0.316	-	0.155	-	0.248	-	0.197	-	0.229	-
	†DeFRCN [33]	0.304	0.626	0.116	0.618	0.166	0.643	0.209	0.614	0.199	0.625
	G-FSDet [39]	0.305	0.695	0.158	0.693	**0.233**	0.702	0.210	0.680	0.227	0.693
	SAE-FSDet [65]	0.324	-	0.157	-	0.191	-	0.205	-	0.219	-
	†TFA [31]	0.130	**0.707**	0.079	**0.707**	0.123	**0.728**	0.130	**0.694**	0.112	**0.709**
	Ours	**0.356**	0.627	**0.190**	0.618	0.231	0.651	**0.220**	0.623	**0.249**	0.630
10	Faster R-CNN [7]	0.000	0.442	0.018	0.432	0.000	0.491	0.000	0.508	0.005	0.468
	FSRW [29]	0.181	0.495	0.090	0.464	0.138	0.445	0.072	0.457	0.120	0.465
	Meta RCNN [63]	0.220	0.494	0.153	0.503	0.203	0.522	0.170	0.523	0.187	0.511
	P-CNN [36]	0.276	0.509	0.189	0.525	0.234	0.521	0.189	0.517	0.222	0.518
	MM-RCNN [37]	0.288	0.501	0.201	0.521	0.250	0.524	0.203	0.521	0.236	0.517
	SAGS&TFS [64]	0.316	-	0.155	-	0.248	-	0.197	-	0.229	-
	†DeFRCN [33]	0.325	0.625	0.177	0.613	0.212	0.634	0.256	0.605	0.243	0.619
	G-FSDet [39]	0.375	0.690	0.207	0.687	0.262	0.711	0.258	0.686	0.276	0.694
	SAE-FSDet [65]	0.371	-	0.174	-	0.284	-	0.227	-	0.264	-
	†TFA [31]	0.150	**0.712**	0.074	**0.706**	0.140	**0.726**	0.134	**0.695**	0.125	**0.710**
	Ours	**0.402**	0.644	**0.248**	0.630	**0.289**	0.664	**0.269**	0.607	**0.302**	0.636
20	Faster R-CNN [7]	0.071	0.458	0.018	0.463	0.112	0.485	0.024	0.452	0.056	0.465
	FSRW [29]	0.220	0.500	0.141	0.435	0.185	0.435	0.122	0.444	0.167	0.454
	Meta RCNN [63]	0.243	0.493	0.205	0.523	0.263	0.537	0.220	0.534	0.233	0.522
	P-CNN [36]	0.296	0.522	0.228	0.516	0.288	0.531	0.257	0.523	0.267	0.523
	MM-RCNN [37]	0.308	0.529	0.239	0.525	0.305	0.536	0.271	0.531	0.281	0.530
	SAGS&TFS [64]	0.402	-	0.238	-	0.361	-	0.277	-	0.320	-
	†DeFRCN [33]	0.352	0.647	0.227	0.615	0.254	0.637	0.314	0.606	0.287	0.626
	G-FSDet [39]	0.398	0.698	0.227	0.682	0.321	0.713	**0.318**	0.677	0.316	0.693
	SAE-FSDet [65]	0.425	-	0.213	-	0.299	-	0.268	-	0.301	-
	†TFA [31]	0.158	**0.710**	0.094	**0.707**	0.109	**0.729**	0.144	**0.700**	0.126	**0.712**
	Ours	**0.430**	0.655	**0.256**	0.633	**0.326**	0.659	0.310	0.606	**0.331**	0.638
30	Faster R-CNN [7]	0.326	0.533	0.257	0.530	0.321	0.536	0.287	0.536	0.298	0.534
	FSRW [29]	0.261	0.515	0.125	0.515	0.213	0.523	0.161	0.521	0.190	0.519
	FSCE [32]	0.240	0.506	0.098	0.492	0.199	0.526	0.145	0.505	0.171	0.507
	Meta RCNN [63]	0.296	0.504	0.144	0.444	0.231	0.452	0.159	0.417	0.208	0.454
	P-CNN [36]	0.281	0.518	0.235	0.532	0.290	0.546	0.253	0.537	0.265	0.533
	MM-RCNN [37]	0.341	0.541	0.272	0.527	**0.345**	0.534	0.302	0.538	0.315	0.535
	†DeFRCN [33]	0.379	0.616	0.253	0.611	0.277	0.636	0.295	0.587	0.301	0.613
	†G-FSDet [39]	0.430	0.698	0.279	0.670	0.312	0.716	0.284	0.678	0.326	0.691
	†TFA [31]	0.168	**0.709**	0.087	**0.707**	0.154	**0.729**	0.128	**0.701**	0.134	**0.712**
	Ours	**0.452**	0.664	**0.291**	0.625	0.339	0.665	**0.343**	0.615	**0.356**	0.642

each class. SAGS&TFS [64] leverages self-adaptive global similarity and a two-way foreground stimulator to enhance performance. G-FSDet [39] integrates global feature alignment and self-adaptive strategies, and achieves amazing performance improvements. SAE-FSDet [65] employs a self-adaptive ensemble mechanism.

More detailed information can be found in Section II.

The results are shown in Tables II and III. We reimplement TFA [31] and DeFRCN [33] with the same training strategy

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TABLE III

COMPARISON OF FSOD DETECTION RESULTS ON NWPU VHR-10.V2 DATASET UNDER 3-, 5-, 10-, AND 20-SHOT SETTINGS AND 2 DIFFERENT BASE/NOVEL CLASS SPLITS. nAP DENOTES THE MAP ON NOVEL CLASSES WHILE bAP DENOTES THE mAP ON BASE CLASSES.† MEANS WE OBTAIN RESULTS BY DIRECTLY RUNNING THE RELEASED CODE

#Shots	Method	Class Split1		Class Split2		Average	
		nAP	bAP	nAP	bAP	nAP	bAP
3	FSRW [29]	0.154	0.831	0.286	0.853	0.220	0.842
	Meta RCNN [63]	0.205	0.870	0.214	0.869	0.210	0.869
	FsDetView [67]	0.246	0.877	0.390	0.881	0.318	0.879
	P-CNN [36]	0.418	0.828	0.393	0.810	0.406	0.819
	†DeFRCN [33]	0.435	0.851	0.325	0.862	0.380	0.857
	G-FSDet [39]	0.491	0.891	0.501	0.900	0.496	0.896
	SAE-FSDet [65]	**0.580**	-	-	-	-	-
	†TFA [31]	0.060	**0.896**	0.079	**0.901**	0.070	**0.899**
	Ours	0.561	0.894	**0.487**	**0.901**	**0.524**	0.898
5	FSRW [29]	0.162	0.828	0.315	0.858	0.239	0.843
	Meta RCNN [63]	0.218	0.857	0.353	0.874	0.286	0.866
	FsDetView [67]	0.296	0.878	0.403	0.893	0.350	0.886
	P-CNN [36]	0.492	0.829	0.461	0.812	0.477	0.821
	†DeFRCN [33]	0.556	0.855	0.370	0.854	0.463	0.855
	G-FSDet [39]	0.544	0.860	0.588	**0.905**	0.575	0.895
	SAE-FSDet [65]	0.594	-	-	-	-	-
	†TFA [31]	0.069	0.896	0.078	0.902	0.074	**0.899**
	Ours	**0.647**	**0.898**	**0.635**	0.898	**0.641**	0.898
10	FSRW [29]	0.240	0.839	0.325	0.855	0.283	0.847
	Meta RCNN [63]	0.270	0.870	0.371	0.876	0.321	0.873
	FsDetView [67]	0.318	0.878	0.451	0.893	0.385	0.886
	P-CNN [36]	0.633	0.831	0.559	0.809	0.596	0.820
	†DeFRCN [33]	0.620	0.863	0.508	0.827	0.564	0.845
	G-FSDet [39]	0.718	0.884	0.670	0.892	0.694	0.888
	SAE-FSDet [65]	0.710	-	-	-	-	-
	†TFA [31]	0.074	**0.896**	0.120	**0.903**	0.097	**0.900**
	Ours	**0.756**	0.892	**0.694**	0.893	**0.725**	0.893
20	FSRW [29]	0.272	0.828	0.339	0.866	0.306	0.847
	Meta RCNN [63]	0.282	0.873	0.394	0.873	0.338	0.873
	FsDetView [67]	0.327	0.878	0.463	0.893	0.395	0.886
	P-CNN [36]	0.668	0.836	0.584	0.812	0.626	0.824
	†DeFRCN [33]	0.675	0.832	0.598	0.810	0.637	0.821
	G-FSDet [39]	0.754	**0.897**	**0.759**	**0.906**	0.757	**0.902**
	SAE-FSDet [65]	**0.851**	-	-	-	-	-
	†TFA [31]	0.082	0.896	0.126	0.903	0.104	0.900
	Ours	0.759	0.889	**0.759**	0.902	**0.759**	0.896

as our method, while the results of other methods are from their original papers.

The detection results on DIOR [60] are presented in Table II. We first compare our method with the baseline model TFA [31]. Since TFA [31] freezes all modules except for the detection head in the fine-tuning stage that maintains most base classes knowledge, it achieves the best detection results of base classes. However, TFA suffers from low performances on novel classes as the fine-tuning strategy significantly limits the learning process. We explore the effect of different parameter fine-tuning strategies in the ablation study similar to G-FSDet [39], but our experiment is more comprehensive. Compared to the baseline TFA, our proposed model obtains consistent performance improvements on novel classes with acceptable performance drops on base classes under all few-shot settings and class splits. The performance improvements are mainly due to the improved fine-tuning stage that makes the best tradeoff between keeping performance on base classes while improving performance on novel classes. More importantly, the text prototypes from CLIP [41] provide more robust supervision signals that help the detection model learn a better feature space. The visual knowledge distillation further boosts the multimodal feature learning and the prompt learning alleviates the domain gap between CLIP [41] and RSIs. As a result, our method averagely obtains 10.4%, 13.7%, 17.7%, 20.5%, and 22.2% mAP gains under all few-shot settings, respectively, of four different novel/base class splits, which are relatively large margins.

Moreover, compared to the other state-of-the-art FSOD methods, our proposed method is superior in most settings. While our method has a simpler framework, it outperforms

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Fig. 2. Examples of novel class detection by different methods under the 10-shot setting in the first class split of the DIOR dataset. The first row shows the ground-truth annotations. The second row illustrates the results of TFA [31] while the bottom row indicates the results of our method. It is shown that TFA tends to ignore novel instances due to the parameter-freezing strategy.

one of the leading meta-learning-based methods named MM-RCNN [37], on both base classes and novel classes. In comparison to G-FSDet [39], the state-of-the-art transfer-learning-based method, our approach excels in novel classes but shows less favorable results for base classes. The reason is that G-FSDet [39] is specially designed to maintain base classes knowledge. In conclusion, the comparison with the baseline and the state-of-the-art model demonstrates the validity of our proposed method. The visualization of detection results of both base classes and novel classes under the 10-shot setting are shown in Fig. 2, which indicates that our method is more effective in alleviating both false detection and missing problem than the baseline.

Results on NWPU VHR-10.v2 [62] are given in Table III. NWPU VHR-10.v2 is a simpler dataset without high intraclass variations compared to DIOR. Therefore, the text prototypes from pretrained CLIP [41] are relatively robust on both base classes and novel classes. As a result, there is less catastrophic forgetting on base classes as shown in Table III. Specifically, our method surpasses the state-of-the-art model G-FSDet [39] by 7%, 10.3%, 3.8%, and 0.4% AP points in from 3- to 20-shot settings, respectively, on novel classes in the first base/novel class split. Moreover, our method obtains 0.3%, 3.8%, and 0.8% more AP points on base classes. The experiment results again verify the effectiveness of our method. The visualized detection results of the 10-shot setting are illustrated in Fig. 3, which also shows the advantage of our method intuitively.

D. Ablation Studies

1) Effects of Different Parameter Fine-Tuning Strategies: In general, the two-stage anchor-based detector can be divided into four parts, i.e., the backbone, RPN, RoI feature extractor, and the detection head. TFA [31] has demonstrated that simply fine-tuning the detection head is effective for FSOD. However, it is demonstrated by several methods [33], [35], [39] that each component is important to be considered whether to be fine-tuned in the second stage. Here we conduct different parameter fine-tuning strategies for each component except for the detection head and the results are reported in Table IV. It is shown that fine-tuning any one component is greatly helpful for improving the detection performances on novel classes. If only one component is fine-tuned, it is interesting that the fine-tuned RPN wins when the number of shots is small, but the fine-tuned backbone performs the best when the number of shots is relatively large. We conjecture that the frozen RPN is biased to the base classes and is prone to treat novel objects as backgrounds. Therefore, the model fails to detect novel objects when there are few of them in the training samples. However, fine-tuning the backbone suffers from severe catastrophic forgetting on base classes, from 9.1% to 10.9% AP drops. This indicates the backbone is the most important one that influences the robustness of image features. According to the table, freezing the backbone while unfreezing others performs the best on novel classes if we treat the backbone and FPN as a whole. In the bottom line of the table, we conduct an additional experiment that only freezes the backbone without FPN and it shows further improved performances on novel classes.

Effects of the Proposed Modules: We conduct extensive ablation experiments on novel classes of the class split 1 of DIOR [60]. The results are introduced in Table V. Specifically, the first row presents the performances of baseline TFA [31]. The subsequent rows first adopt the parameter fine-tuning strategy and then combine the three different modules proposed only for training. We first generate class prototypes

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TABLE IV

ABLATION STUDY ON THE EFFECTS OF DIFFERENT PARAMETER FINE-TUNING STRATEGIES. THE EXPERIMENTS ARE CONDUCTED IN CLASS SPLIT 1 OF DIOR. THERE ARE THREE COMPONENTS HERE TO BE FINE-TUNED. RoI FEAT. INDICATES THE SHARED RoI FEATURE EXTRACTOR

Method	Fine-tune				3shot		5shot		10shot		20shot		30shot	
	Backbone ResNet	Backbone FPN	RPN	RoI Feat.	nAP	bAP	nAP	bAP	nAP	bAP	nAP	bAP	nAP	bAP
TFA	✗	✗	✗	✗	0.138	0.706	0.130	**0.707**	0.150	**0.712**	0.158	**0.710**	0.168	**0.709**
Fine-tuning Strategy	✔	✔	✗	✗	0.242	0.615	0.286	0.607	0.354	0.616	0.382	0.620	0.417	0.630
	✗	✗	✔	✗	0.291	**0.710**	0.310	0.702	0.335	0.695	0.356	0.703	0.363	0.703
	✗	✗	✗	✔	0.170	0.663	0.180	0.665	0.205	0.673	0.214	0.686	0.226	0.684
	✔	✔	✔	✗	0.240	0.611	0.281	0.608	0.355	0.616	0.385	0.619	0.422	0.629
	✔	✔	✗	✔	0.260	0.601	0.296	0.599	0.358	0.607	0.394	0.615	0.432	0.625
	✗	✗	✔	✔	0.301	0.661	0.320	0.663	0.371	0.671	0.405	0.687	0.430	0.679
	✔	✔	✔	✔	0.258	0.597	0.297	0.599	0.360	0.609	0.393	0.614	0.423	0.626
	✗	✔	✔	✔	**0.314**	0.646	**0.331**	0.638	**0.388**	0.651	**0.417**	0.672	**0.442**	0.674

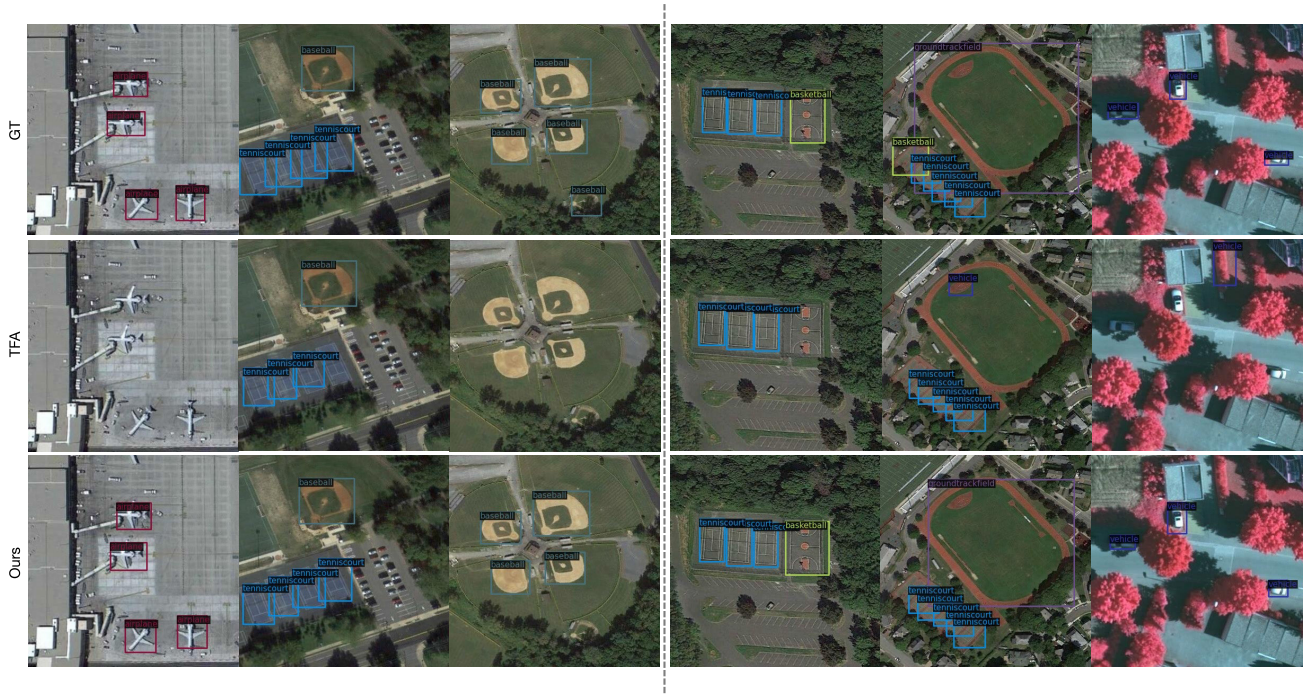


Fig. 3. Examples of novel class detection by different methods under the 10-shot setting in the two class splits of NWPU VHR-10.v2 dataset (on each side of the dotted line). The first row shows the ground-truth annotations. The second row illustrates the results of TFA [31] while the bottom row indicates the results of our method.

with the CLIP [41] text encoder. The prompts are fixed as a template "a centered satellite photograph of " followed by the class names. According to the table, there seems a slight improvement (such as 0.5% AP points increment under the 3-shot setting and the 5-shot setting). We believe one reason is the gap between the visual semantic space of the detection head and the language semantic space of CLIP. CLIP was pretrained on the image–text pairs which is irrelative to the object detection task. Therefore, we adopt the CLIP visual encoder as a teacher to directly guide the optimization of the embedding features from the detection head, which boosts the alignment between the features from the detection model and CLIP. As a result, it leads to consistent improvements under all settings. Another factor hindering the detection performance is that RSIs contribute little to CLIP pretraining and there is a domain gap between the natural images and the RSIs. Consequently, when we add the prompt tuning to adapt CLIP to the remote sensing scenario, the detection performances continue to increase especially when the number of shots is small. To summarize, the results in Table V verify the effectiveness of each module proposed in our method. Moreover, Fig. 4 illustrates t-SNE visualization results of object features. It shows that although CLIP prototypes do not closely align with the true centroids, the feature space learned under the guidance of CLIP prototypes performs better than that learned from a limited set of visual samples.

We present detailed experimental results for each novel class in Table VI on class split 1 of the DIOR [60] dataset under the 3-shot setting. The proposed method in this article significantly improves the FSOD performance on most of novel classes except for the "bridge." The reason may be that the training samples of "bridge" are similar and the model overfits them.

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TABLE V

ABLATION STUDY ON EFFECTS OF DIFFERENT MODULES IN THE PROPOSED METHOD. THE EXPERIMENTS ARE CONDUCTED ON NOVEL CLASSES IN CLASS SPLIT 1 OF DIOR (REPORTED AS NAP). TPS, VKD, AND PL INDICATE TEXT PROTOTYPE SUPERVISION, VISUAL KNOWLEDGE DISTILLATION, AND PROMPT TUNING, RESPECTIVELY

Method	TPS	VKD	PL	#Shots				
				3	5	10	20	30
TFA [31]	✗	✗	✗	0.138	0.130	0.150	0.158	0.168
Fine-tuning	✗	✗	✗	0.314	0.331	0.388	0.417	0.442
Ours	✔	✗	✗	0.319	0.336	0.394	0.421	0.451
	✔	✔	✗	0.330	0.347	0.395	0.426	**0.454**
	✔	✔	✔	**0.342**	**0.356**	**0.402**	**0.430**	0.452

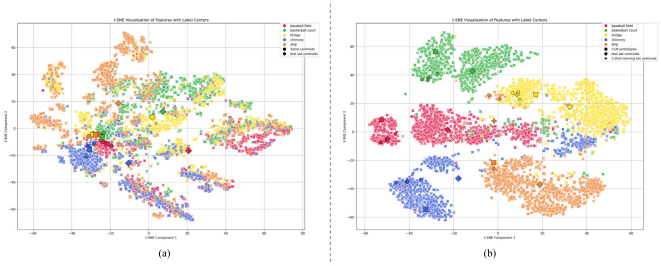


Fig. 4. t-SNE visualization of novel class object features learned by the baseline TFA [31] and our method. Results are obtained under the 10-shot setting of split 1 of the DIOR dataset. We randomly choose 500 objects for each novel class from the test set.

TABLE VI

ABLATION STUDY ON EFFECTS OF DIFFERENT MODULES IN THE PROPOSED METHOD FOR EACH NOVEL CLASS. THE EXPERIMENTS ARE CONDUCTED ON NOVEL CLASSES IN CLASS SPLIT 1 OF DIOR UNDER THE 3-SHOT SETTING (NAP). TPS, VKD, AND PL INDICATE TEXT PROTOTYPE SUPERVISION, VISUAL KNOWLEDGE DISTILLATION, AND PROMPT TUNING, RESPECTIVELY

Method	TPS	VKD	PL	baseball field	basketball court	bridge	chimney	ship
TFA [31]	✗	✗	✗	0.000	0.093	**0.214**	0.303	0.081
Fine-tuning	✗	✗	✗	0.346	0.425	0.142	0.567	**0.091**
Ours	✔	✗	✗	0.350	0.447	0.131	0.575	**0.091**
	✔	✔	✗	0.360	**0.458**	0.148	0.591	**0.091**
	✔	✔	✔	**0.413**	0.451	0.144	**0.609**	**0.091**

Moreover, the model performs poorly on the "ship" due to its high intraclass variance. It needs a more robust text representation. Comparatively, the model performs well on novel classes with lower intraclass variance, i.e., "baseball field" and "basketball field." Overall, the proposed method can effectively improve the FSOD performance.

2) Choice of the Weight of $\mathcal{L}_{vkd}$: The weight of $\mathcal{L}_{\text{vkd}}$ controls the balance between the text prototype supervision and visual knowledge distillation. We conduct an ablation study to choose the best weight of $\mathcal{L}_{\text{vkd}}$ on DIOR [60] class split 1, the results are shown in Table VII. It can be observed that when the number of shots is low (such as 3- and 5-shot), a larger weight of $\mathcal{L}_{\text{vkd}}$ obtains relatively higher performance on novel classes. However, with the number of shots increasing, the performance on novel classes of larger weights is lower than that of smaller weights. The reason may be that the varied visual features increase the difficulty of visual knowledge distillation when the number of shots is relatively large. Moreover, the performance on base classes decreases with the increase in weight. We believe that the visual features from CLIP are less robust than those learned from a large number of training samples.

TABLE VII

ABLATION STUDY OF THE WEIGHT OF $\mathcal{L}_{\text{VKD}}$ ON DIOR CLASS SPLIT 1, WITH THE RESULTS REPORTED AS NAPS FOR THE NOVEL CLASSES

Weight of $\mathcal{L}_{vkd}$	3-shot		5-shot		10-shot		20-shot		30-shot	
	nAP	bAP	nAP	bAP	nAP	bAP	nAP	bAP	nAP	bAP
0.1	**0.342**	**0.627**	0.356	**0.627**	**0.402**	**0.644**	**0.430**	**0.655**	**0.452**	**0.664**
0.5	**0.342**	0.618	**0.362**	0.615	0.391	0.629	0.423	0.649	0.446	0.655
1.0	**0.342**	0.613	0.361	0.604	0.386	0.621	0.419	0.639	0.443	0.649

TABLE VIII

ABLATION STUDY OF THE HYPERPARAMETER M IN PROMPT LEARNING ON DIOR CLASS SPLIT 1, WITH THE RESULTS REPORTED AS NAPS FOR THE NOVEL CLASSES

M	3-shot		5-shot		10-shot		20-shot		30-shot	
	nAP	bAP	nAP	bAP	nAP	bAP	nAP	bAP	nAP	bAP
4	0.338	0.621	0.349	0.623	0.398	0.636	0.429	**0.659**	0.454	0.657
8	0.338	**0.639**	0.355	0.624	0.402	0.640	0.426	**0.659**	0.454	0.663
16	**0.342**	0.627	**0.356**	0.627	0.402	**0.644**	0.430	0.655	0.452	**0.664**
24	0.333	0.627	0.354	**0.628**	**0.403**	0.639	**0.431**	0.657	**0.456**	0.655

TABLE IX

COMPARISON AMONG DIFFERENT PRETRAINED VLMS ON DIOR DATASET SPLIT 1 WITH A FIXED PROMPT

VLM	3-shot		5-shot		10-shot		20-shot		30-shot	
	nAP	bAP	nAP	bAP	nAP	bAP	nAP	bAP	nAP	bAP
CLIP [41]	0.330	0.629	0.347	**0.627**	**0.395**	0.638	**0.426**	**0.656**	**0.454**	0.656
BLIP [68]	**0.340**	**0.634**	0.354	0.624	0.391	0.630	0.422	0.645	0.445	**0.660**
FLAVA [69]	0.332	**0.634**	**0.357**	0.624	**0.395**	0.636	0.420	0.646	0.446	0.659

3) Hyperparameter Sweep in Prompt Learning: The number of prompt vectors M in prompt learning affects the representation ability of each text prompt. We conduct the hyperparameter sweep of different numbers on DIOR [60] class split 1, and results are presented in Table VIII. From this table, we can see that with the increase of M, the performance generally increases slightly. It indicates the more prompt vectors it has, the better it can represent different classes. However, the performance varies when comparing $M = 16$ and $M = 24$. This demonstrates that too many vectors are hard to learn given low-shot samples. To achieve an optimal balance between performance and cost, we finally set M to 16.

4) Comparison Among Different Pretrained VLMs: We further import other pretrained VLMs to compare them with CLIP [41]. The VLMs should have independent texts and visual encoders to extract text prototypes and ground truth visual features in different steps. Specifically, we choose two well-known VLMs named BLIP [68] and FLAVA [69]. We use the simplest template "a centered satellite photo of {class name}." for all VLMs, and the experimental results are given in Table IX. It can be seen that the performance gaps among the VLMs are relatively small with the same

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prompts. The reason may be that they all have powerful multimodal feature extraction abilities and the simple template is hard to distinguish them. However, the results still verify the robustness of our method to different pretrained VLMs.

VI. Conclusion

In this work, we propose to utilize natural language to generate class prototypes with the help of pretrained VLMs rather than only using limited visual instances to achieve better FSOD performance. Specifically, we leverage a pretrained VLM CLIP [41] as the teacher and adopt both language knowledge distillation and visual knowledge distillation. Moreover, as CLIP [41] is pretrained mostly on natural image–text pairs, we utilize prompt tuning as an auxiliary optimization method for domain adaptation. Extensive experiments on two popular remote sensing FSOD datasets verify the effectiveness of the proposed method.

A majority of current FSOD methods try to mine as much information as possible from limited visual samples. However, the detection performance is still far from the sufficiently supervised training due to the lack of diversity. Our method attempts to import general knowledge from a pretrained VLM to FSOD, which is still a preliminary exploration. In the future, we plan to explore more efficient ways to exploit VLMs such as advanced domain adaptation and visual prompt tuning. Moreover, generative models can also be used to synthesize more training samples.

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